Base Editing in Human Cells

Abstract

Base editing converts one target nucleotide into another without creating double-strand breaks. In this study we compared cytosine and adenine base editors across multiple cell types and quantified on-target editing, indel formation, and transcriptome-wide off-target effects.

Introduction

Programmable nucleases have transformed genome engineering, yet their reliance on double-strand breaks creates a mixture of insertions and deletions. Base editors avoid this by coupling a catalytically impaired Cas9 to a deaminase, enabling direct chemical conversion of a target base.

Results

We observed efficient conversion at all tested loci. Editing efficiency ranged from twenty to sixty percent depending on the editor and the target motif. Indel frequencies remained below two percent. RNA sequencing revealed no significant transcriptome-wide changes.

Discussion

These results support the use of base editors for precise nucleotide substitution. Careful target selection and delivery optimization remain necessary before therapeutic applications, and long-term safety requires further evaluation.